

Trustworthy LLM-Assisted Venture Screening: Auditing Expert Alignment, Evidence Grounding, and Cross-Domain Calibration

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Abstract

Large language models are moving into early-stage investment workflows, screening deal flow and shaping which ventures reach human decision-makers – yet the scores they produce are rarely audited as measurement instruments before informing capital allocation. We present a validity audit of an evidence-grounded, nine-stage LLM pipeline that scores venture proposals and issues GO/NO-GO screening recommendations, evaluated against two complementary ground truths: working screening assessments from an operating investment firm’s deal flow (12 real venture proposals, independently scored by a three-rater expert panel), and objective, all-or-nothing funding outcomes for 141 real Kickstarter campaigns the pipeline was never tuned on. We decompose validity into six independently testable targets and audit each. The pipeline’s score carries genuine, transferable investment signal: it correlates significantly with real out-of-domain funding outcomes (Spearman $\rho = 0.366$, Holm $p < 0.001$; ROC-AUC = 0.764) and grounds its claims largely in source evidence (81.8% lenient groundedness). Yet expert-alignment validity fails decisively. The panel is highly reliable (ICC(2,1) 0.79–0.88; ICC(2,k) 0.92–0.95), establishing a human agreement ceiling against which the pipeline falls well short: its rank agreement with expert consensus ($\rho \approx 0.05$ –0.17) sits below every individual rater’s leave-one-out agreement with the panel ($\rho \approx 0.80$ –0.95) on every scorable target. The audit further isolates where trust breaks down operationally: the decision threshold calibrated on venture deal flow flags zero fundable campaigns out-of-domain (0% recall) until cross-validated recalibration restores 58.3%, and the pipeline’s confidence output does not track whether its recommendations are right. Useful outcome signal can therefore coexist with a decisive expert-alignment failure – a combination neither an outcome-free audit nor an outcome-based one would have revealed alone. We derive concrete governance requirements from the specific failure modes the audit exposes, supporting audited LLM screening scores for human-supervised investment triage, not autonomous investment decisions or general venture-success prediction.

CCS Concepts

• **Computing methodologies** → **Natural language processing**;
• **Information systems** → *Evaluation of retrieval results*; • **Applied computing** → *Economics*.

Keywords

trustworthy AI, large language models, venture screening, financial decision support, AI governance, model validation, calibration, groundedness

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1 Introduction

Large language models are entering early-stage investment workflows and shaping which ventures reach human decision-makers, but the key question is not whether these models produce convincing reports – it is whether the judgments behind them are grounded, calibrated, and governable before they inform capital allocation. We study this in venture screening, where investors assess teams, markets, strategy, feasibility, and valuation under substantial uncertainty [10]. Because these judgments shape which ventures receive attention or funding, model outputs must also be traceable, reliable, and suitable for human oversight.

Venture proposals combine incomplete evidence across narrative text, financial projections, tables, figures, IP claims, team profiles, timelines, and regulatory plans, making screening a mixed-format financial reasoning task [5, 10]. Prior work frames innovation finance as a domain-specific classification and recommendation problem [12], but strong extraction performance does not guarantee reliable reasoning, calibrated recommendations, or responsible decisions [29]. Trustworthy financial AI therefore requires evidence grounding, traceability, and governance alongside analytical capability.

When long-term venture outcomes are unavailable, outcome-free auditing provides a proximal, not final, validity test for trustworthy financial AI. Outcomes may take years to emerge and depend on capital, execution, market timing, regulation, macroeconomic shocks, and post-investment support [10]. VC-specific LLM benchmarks also face sparse private signals, privacy constraints, and uncertain long-horizon labels [3]. Before such labels mature, screening scores should align with real screening judgments, preserve rankings, remain traceable to source evidence, use calibrated thresholds, and respond appropriately to evidence changes [9, 15]. Passing such tests supports constrained human review, not venture-success prediction.

Recent work motivates a broader trustworthy-AI audit of financial decision support. FAITH shows LLMs can produce unsupported numerical claims even when relevant tables are available [33], while investment models may retain initial preferences despite strong counter-evidence [16]. Financial agents should therefore be evaluated through intermediate behavior and human collaboration, not final accuracy alone [25]. Following RAG and factuality research [7, 22], we audit source support and generated judgments separately, then compare scores against real screening decisions, observed outcomes, and rubric-targeted perturbations – directly evaluating explainability, robustness, and human oversight in responsible financial AI.

The closest prior startup-evaluation system is DIALECTIC, which structures startup facts, question trees, pro/con reasoning, critique, and numeric investment scores [1]. Our contribution is not another scoring architecture, but a trustworthy-AI audit of the score itself. We develop a nine-stage, provenance-preserving pipeline that scores venture proposals and issues GO/NO-GO screening recommendations, evaluated against two ground truths: an operating investment firm’s deal flow (12 real venture proposals) and 141 Kickstarter campaigns with observed, objective funding outcomes. Concretely, the pipeline’s score carries real, transferable signal – it correlates significantly with funding outcomes in a domain it was never tuned on (Spearman $\rho = 0.366$, Holm $p < 0.001$; ROC-AUC = 0.764) – yet the same audit exposes where that trust breaks down: its decision threshold transfers with 0% recall until recalibration restores 58.3%, and its confidence output does not track whether recommendations are right. This reframes startup evaluation from whether an LLM can generate a plausible score to what validation and governance are required before that score can support accountable, human-supervised triage.

Our central research question is:

Can an evidence-grounded LLM produce reliable, traceable, and calibrated venture-screening scores for auditable human-supervised review?

We address four research questions:

- RQ1.** How well do LLM scores match real screening judgments?
- RQ2.** Are factual claims in the generated assessments supported by the source proposals?
- RQ3.** Do the scores relate to observed funding outcomes, and do decision thresholds work across domains?
- RQ4.** Do scores change as expected when rubric-relevant evidence is added or removed?

We make five contributions:

- (1) We develop an auditable, nine-stage pipeline that converts mixed-format venture proposals into source-linked facts, structured evidence packets, generated assessments, rubric-based scores, and screening recommendations.
- (2) We present a multidimensional audit framework for trustworthy financial AI that separately evaluates rank preservation, scale bias, claim grounding, evidence responsiveness, outcome alignment, and threshold transfer.
- (3) We establish a human–human reliability ceiling from a three-rater expert panel (ICC(2,1) 0.79–0.88) and show that the pipeline’s agreement with expert consensus falls below every individual rater’s leave-one-out agreement with the panel on every scorable target.
- (4) We evaluate the pipeline on 153 real-world screening cases: 12 venture proposals assessed by a three-rater expert panel and 141 Kickstarter campaigns with observed funding outcomes. The results reveal a significant out-of-domain outcome signal alongside a decisive expert-alignment failure and specific, measurable weaknesses in grounding, calibration, scale agreement, and confidence reliability.
- (5) We show that score validity and decision-threshold validity are distinct: a score can retain useful out-of-domain discrimination while its original decision threshold fails completely. Cross-validated recalibration restores substantial

recall, while nested validation exposes overfitting and data-leakage risks in apparently simple score adjustments.

These findings support LLM-generated screening scores for auditable, human-supervised triage, not autonomous investment decisions or general venture-success prediction.

2 Related Work

2.1 Financial LLMs and Trustworthy Venture Screening

Financial LLM research has shifted from domain adaptation to task-specific evaluation: BloombergGPT, FinGPT, and PIXIU show the value of financial pretraining and instruction resources [28, 30, 31]; FinEval, CFBenchmark, and FinBen show uneven performance across financial tasks [18, 29, 32]; FinanceRAG and Multi-Reranker highlight retrieval and context management in financial QA [8, 17]. None evaluate whether an LLM-generated investment score is valid, traceable, and calibrated for human oversight and governance.

Venture screening spans teams, business models, valuation, deal structure, and post-investment risk [10]. Automating Venture Capital and VCBench predict founder or startup success from sparse, private signals [3, 23], while DIALECTIC structures startup facts, reasoning, critique, and investment scores [1] – all optimizing predictive accuracy on outcome labels. We instead audit whether a deployed score is *valid*: whether it preserves real screening rankings, remains grounded, retains outcome signal, and uses thresholds that transfer across domains – a measurement and governance question distinct from predictive performance.

2.2 Grounding and Trustworthy Financial Decision Support

Trustworthy financial AI needs calibrated, robust, and traceable recommendations with human oversight. Investment-risk assessments may vary across country and gender attributes [2]; investment preferences may persist despite counter-evidence [16]; and agents should be evaluated through intermediate behavior and human collaboration, not final accuracy alone [25]. Transparency, prompts, trust, accountability, and task difficulty further shape LLM-assisted decisions [6, 19]. We therefore audit the scoring system’s behavior, evidence, and deployment limits – including fairness and privacy exposure – rather than the surface fluency of its reports.

Evidence grounding is also essential. RAGAS and ARES separate context quality from answer faithfulness [7, 24], while FActScore and SelfCheckGPT detect unsupported claims [21, 22]; FAITH shows financial hallucination rises as reasoning grows more complex [33]. Since investment reports mix verifiable claims with professional judgments, we check source claims against the proposal while evaluating judgments through rank preservation, rubric consistency, and responsiveness to evidence.

2.3 Human-AI Agreement and Measurement Validity

Venture screening is a judgment task, so evaluation should compare LLM scores against real screening decisions, not a fixed answer key. Evaluation choices can produce conflicting conclusions about model performance [15]. LLM-as-judge methods support

rubric-based scoring but may introduce evaluator bias: G-Eval uses guided criteria [20], MT-Bench and Chatbot Arena reveal preference biases [4, 34], and Prometheus models emphasize explicit rubrics [13, 14]. We therefore assess rank agreement, absolute agreement, calibration, and dimension reliability separately, rather than collapsing validity into a single accuracy number.

Prior work thus covers financial LLMs, venture prediction, structured scoring, grounding, and LLM evaluation, but no unified audit jointly tests whether a deployed screening score preserves real-world rankings, remains unbiased on an absolute scale, stays evidence-grounded, responds correctly to evidence changes, retains signal on real outcomes, and uses thresholds that transfer across domains. We address this as a trustworthy financial-AI measurement, calibration, and governance problem, surfacing leakage risks in score adjustment and threshold selection along the way.

3 System Architecture

3.1 Design Philosophy

We frame the system as a measurement instrument, not a success predictor: venture outcomes take years and are confounded by capital, execution, and timing, so we instead ask whether scores agree with real screening judgments, stay traceable to evidence, and respond predictably as evidence changes. This motivates a *provenance-preserving* design – each stage persists an auditable artifact to disk, so scores trace back to specific extracted facts, following FActScore’s atomic-fact decomposition and RAGAS/ARES’s separation of context quality from answer faithfulness (§2).

We combine two validation philosophies: *outcome-free auditing* against the firm’s screening judgments (12 proposals), since real outcomes are not yet available, and *outcome-based validation* against 141 resolved Kickstarter outcomes, an objective ground truth the former cannot provide. Neither alone suffices – single-rater agreement can reflect one reviewer’s idiosyncrasies, and outcome correlation alone says nothing about grounding – so §6 and §7 report both together.

3.2 Formal Task Definition

Let $H_i = \{h_{\text{team}}, h_{\text{objective}}, h_{\text{strategy}}, h_{\text{advantages}}, h_{\text{feasibility}}, r_i\}$ denote the firm’s assessment of proposal i across five dimensions – team, objectives, strategy, advantages, and feasibility – each scored 1–5, plus an overall ranking r_i scored 1–20; all six are normalized to $[0, 1]$. The system produces the analogous

$$\hat{H}_i = \{\hat{h}_{\text{team}}, \hat{h}_{\text{objective}}, \hat{h}_{\text{strategy}}, \hat{h}_{\text{advantages}}, \hat{h}_{\text{feasibility}}, \hat{r}_i, \hat{c}_i, \hat{y}_i\} \quad (1)$$

where $\hat{r}_i, \hat{c}_i \in [0, 1]$ are the overall score and confidence, and $\hat{y}_i \in \{\text{GO}, \text{HOLD}, \text{NO-GO}\}$ is the screening recommendation: GO advances a proposal toward closer human review and a funding decision, not to funding itself.¹

We decompose validity into six targets: ordinal (rank preservation, Spearman ρ), scale (unbiased values, ICC), evidence (claim

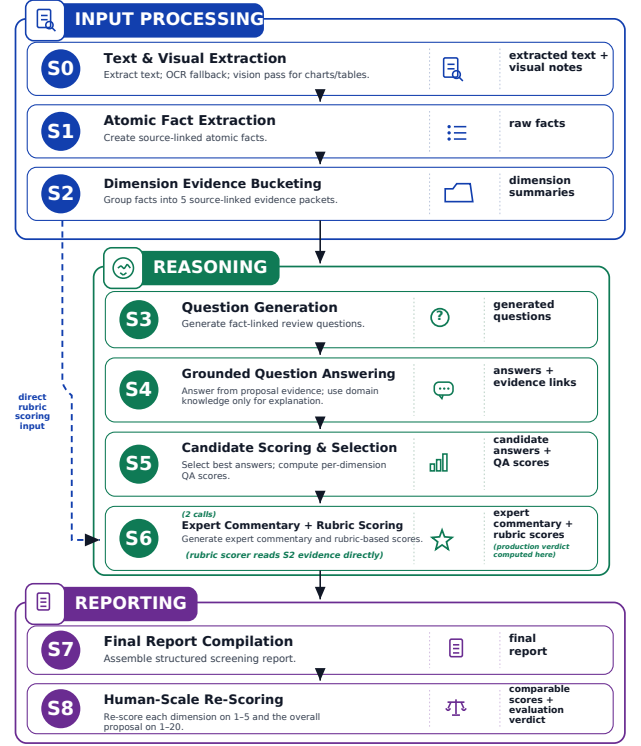


Figure 1: The nine-stage pipeline. Each stage persists an auditable intermediate artifact (raw facts, dimension summaries, generated questions, candidate answers, expert commentary, rubric scores), enabling every final score to be traced to specific source evidence.

grounding, strict/lenient groundedness), responsiveness (movement under prompt-coupled evidence injection), outcome alignment (agreement with real funding outcomes, since screening judgment alone cannot establish external validity), and threshold generalization (whether a decision rule transfers across domains). §6 and §6.3 report each in turn.

3.3 Pipeline Overview

The pipeline decomposes screening into nine sequential stages, each persisting a disk artifact consumed by the next (Fig. 1): **S0** text extraction, with OCR fallback and a vision-model pass for charts and tables; **S1** atomic, source-linked fact extraction; **S2** bucketing facts into five dimensions with grounded evidence packets; **S3** fact-linked question generation; **S4** question answering restricted to proposal facts; **S5** candidate scoring, selection, and per-dimension QA scoring (§3.4); **S6** expert commentary and the investment-rubric score (§3.4); **S7** final report compilation; and **S8** re-scoring on the human 1–5/1–20 scale for comparability.

Stages 6 and 8 can run per-provider for inter-rater comparison; all results here use a single provider (OpenAI gpt-4o-mini), leaving cross-provider agreement open (§8). Despite this granularity, a full run – ~156 LLM calls – completes in under 30 minutes after tuning, down from over 10 hours unoptimized.

¹Internal dimension innovation is reported as advantages.

3.4 Investment Rubric and Verdict Rule

Stage 6 makes two calls: qualitative commentary (temperature=0.25, forbidden from outputting numeric scores) and a deterministic investment-scoring call (temperature=0, seed=42) that reads Stage 2’s evidence directly, bypassing the QA-answer pipeline, instructed: “Score each dimension 0.0–1.0 based *ONLY* on the proposal facts provided. Be discriminating: most proposals should land in 0.40–0.70. Scores above 0.80 require strong, specific evidence.” This counters LLM judges’ tendency to compress scores toward a narrow middle range regardless of true quality [27]; absolute scores land in 0.4–0.7 across every dataset (§6, §8). Concretely: any product described as “AI-powered” or “data-driven” is capped at 0.42–0.56 on innovation regardless of sophistication, unless the AI is merely a design tool for an underlying physical/biological product commercialized on its own merits.

Each dimension’s investment score blends with its QA-quality score (§3.3) at 0.90/0.10:

$$\text{blended}_d = 0.90 \cdot \text{invest_score}_d + 0.10 \cdot \text{qa_score}_d \quad (2)$$

$\hat{r}_i$ re-applies the same weights used at QA-scoring ($w_{\text{team,objectives,strategy}} = 1.00$, $w_{\text{advantages}} = 1.10$, $w_{\text{feasibility}} = 1.20$):

$$\hat{r}_i = \frac{\sum_d w_d \cdot \text{blended}_d}{\sum_d w_d} \quad (3)$$

Confidence $\hat{c}_i$ carries over unchanged from Stage 5. Letting $\hat{h}_{\min} = \min(\hat{h}_{\text{advantages}}, \hat{h}_{\text{feasibility}})$, the verdict is:

$$\text{verdict}(\hat{r}_i, \hat{c}_i) = \begin{cases} \text{GO} & \hat{r}_i \geq 0.725 \vee \hat{c}_i \geq 0.811 \\ \text{NO-GO} & \hat{r}_i < 0.40 \vee \hat{h}_{\min} < 0.30 \\ \text{HOLD} & \text{otherwise} \end{cases} \quad (4)$$

HOLD defers the proposal for further human review rather than advancing or rejecting it; for the binary evaluations in §6, HOLD and NO-GO are both treated as non-GO. Within this dataset NO-GO is in practice unreachable: the rubric’s 0.40–0.70 target band keeps every observed blended dimension score 0.21 or more above the NO-GO cutoffs (§8), so the verdicts realized here are effectively a two-way GO/HOLD split.

INVEST_WEIGHT=0.90 is the production configuration. A post-hoc sweep of $\lambda \in [0, 1]$ against the expert-panel consensus shows Spearman correlation remaining near zero across the entire range (peak $\rho = 0.096$ at $\lambda = 0.95$), so no weight in this range produces meaningful alignment with expert judgment. We therefore report INVEST_WEIGHT=0.90 as a documented heuristic choice rather than an empirically optimized one, and do not re-tune it against this data to avoid re-fitting noise on the same $n = 12$ sample (§8).

4 Datasets

4.1 Expert-Scored Venture Proposals

Twelve real, Chinese-language venture proposals, drawn from an operating investment firm’s evaluation pipeline and spanning biotech, medtech, aerospace, automotive AI hardware, semiconductor, and education domains, were scored by the firm’s expert reviewer on five 1–5 dimension scales plus a 1–20 overall ranking ExpertsEvaluations.xlsx – working screening assessments from live deal flow, not annotations produced for this study. We treat this file as an

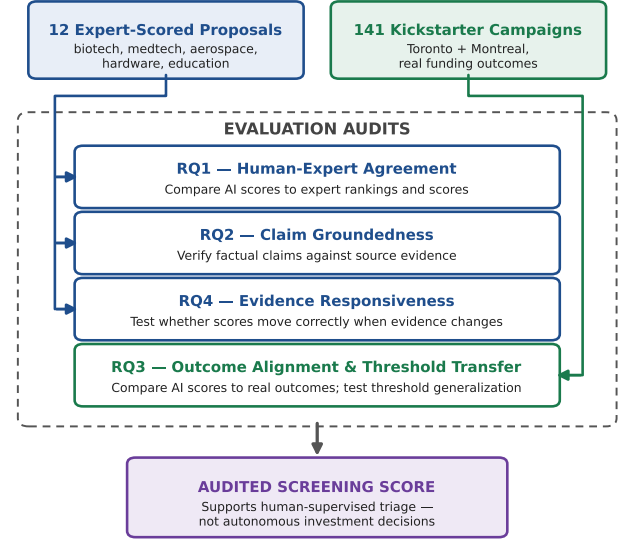


Figure 2: The evaluation program: four parallel, independent audits. RQ1, RQ2, and RQ4 draw on the 12 expert-scored proposals; RQ3 draws on the 141 Kickstarter campaigns with real funding outcomes. All four converge on a single question – whether the score can support accountable, human-supervised triage.

operational reference: no second rater exists, and no held-out test set was constructed.

4.2 Kickstarter Real-Outcome Dataset

141 real Kickstarter campaigns (104 Toronto, 37 Montreal) provide an objective, binary success label. Kickstarter uses all-or-nothing funding: a campaign either reaches 100% of its goal and gets fully funded, or receives nothing. This makes funded% ≥ 100 a clear, fact-based label, unlike our expert dataset, where “success” depends on one person’s subjective judgment (dataset_info.xlsx). All 141 campaigns were scored by the identical, unmodified pipeline (§4.1) – a genuine out-of-domain test, since the pipeline was designed and tuned entirely on VC-style proposals.

5 Evaluation Methodology

We organize validation around the four research questions from §3.2 (Fig. 2), each evaluated on its appropriate dataset (Table 1); §5.1–§5.4 detail each in turn.

5.1 Human-Expert Agreement (RQ1)

We compare AI and human scores using six metrics: weighted Cohen’s κ on scores binned to a 5-point grid, with bootstrap 95% CIs (noisy here, as AI scores concentrate in few bins); ICC(2,1) [26] for absolute agreement on the overall ranking; Spearman ρ on raw $[0, 1]$ scores for the five dimensions plus overall – the primary ranking metric, with these tests Holm-corrected as one family [11]; a paired mean-difference test, $\text{mean}(\hat{r}_i - r_i)$, for scale offset (Wilcoxon

Table 1: Evaluation methodology summary

RQ	Target	Metric(s)	<i>n</i>
RQ1	Rank/scale agree- ment	κ , ICC, Spearman, concordance	12
RQ2	Claim grounding	Strict/lenient groundedness	12
RQ3	Outcome align- ment	Spearman, ROC-AUC, CV thresh- old	141
RQ4	Responsiveness	PLUS/MINUS pass rate	12

as sensitivity check; these bias tests form a second Holm family); pairwise rank concordance (fraction of firm-ordered pairs the AI orders identically); and verdict classification comparing $\hat{y}_i$ to the firm’s funding decision.

At $n = 12$, 80% power at $\alpha = 0.05$ requires $|\rho| \approx 0.73$, so we treat single-dataset correlation results as exploratory, not confirmatory.

5.2 Claim Groundedness and Content Sensitivity (RQ2, RQ4)

Statements are classified as verifiable facts, numerical inferences, professional inferences, or evaluator judgments; only the first two are checkable source claims and count toward groundedness. Each eligible claim is supported, partially supported, or unsupported:

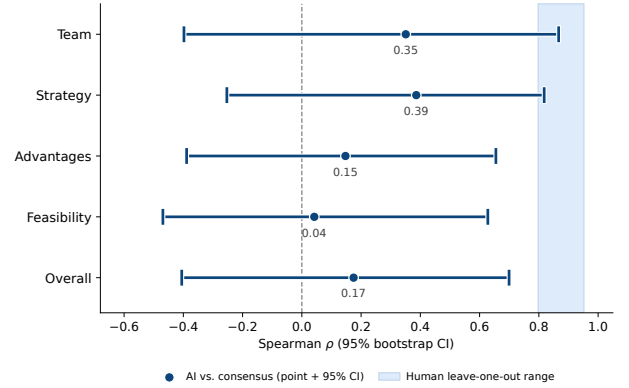
$$G_{\text{strict}} = \frac{N_{\text{supported}}}{N_{\text{eligible}}}, \quad G_{\text{lenient}} = \frac{N_{\text{supported}} + 0.5 \cdot N_{\text{partial}}}{N_{\text{eligible}}} \quad (5)$$

For RQ4, PLUS/MINUS variants perturb Strategy, Objectives, and Advantages: PLUS prepends a fixed favorable-evidence block and should raise scores; MINUS regex-removes narrative lines matching target keywords, leaving Q&A evidence intact, and should lower them. Being prompt-coupled, this measures generation–scoring consistency under rubric-targeted edits, not real-world evidentiary responsiveness.

5.3 Real-World Outcome Validation and Threshold Transfer (RQ3)

We correlate the AI’s overall score with real funded% using Spearman ρ – separately for Toronto, Montreal, and combined, since funding percentage is extremely right-skewed – and repeat this per dimension. We then classify the AI’s GO verdict as a prediction of funded% ≥ 100 , reporting accuracy, precision, recall, and F_1 at the production threshold (§3.4).

Because only 24 of 141 campaigns succeeded, a single train/test split would leave as few as 7 successes in the test set – too few for stable estimates. We instead use stratified 5-fold cross-validation (seed= 42): each fold holds a proportional ($\sim 17\%$) share of successes, the other four grid-search the F_1 -maximizing (τ_r, τ_c) pair over production’s OR-logic shape, and that threshold is applied only to the held-out fold – so every campaign is scored by a threshold that never saw it. We optimize F_1 because at a 17% success rate, always predicting failure already scores 83% accuracy while identifying no

AI agreement with expert consensus falls below the human reliability ceiling

Objective omitted: panel consensus is constant across all 12 proposals (Spearman undefined); reported via Gwet’s AC1 in the text.

Figure 3: AI-versus-consensus Spearman correlations against the human leave-one-out reliability range. Point estimates fall below the human range on every scorable dimension; confidence-interval upper bounds for Team and Strategy partially reach into the human range, reflecting the small sample ($n = 12$). Objective is omitted – consensus is constant across all 12 proposals, so Spearman is undefined; see Gwet’s AC1 (§6.1) for that dimension’s reliability.

true successes. We report three scenarios: the unmodified production threshold; an in-sample fit (overfitting reference only); and the cross-validated result.

5.4 Score Improvement and Robustness Checks

Since RQ3 finds team and objectives carry no significant Kickstarter signal while feasibility, strategy, and advantages do, we compare three score variants via the same 5-fold CV (no confidence OR-gate, isolating score quality): **A**, the production score; **B**, a naive composite of feasibility, strategy, and advantages; and **C**, an L2-regularized logistic regression ($C = 1.0$) fit within each fold across all five dimensions. We report threshold-free ROC-AUC and Average Precision.

Method B’s dimension choice was itself leakage-prone – selected via correlations on the full dataset, a softer form of the problem threshold calibration guards against. We therefore (1) bootstrap each pairwise AUC difference 10,000 times, calling a gap significant only if its 95% CI excludes zero, and (2) fully nest Method B, re-deriving both dimension selection and threshold within each training fold, so no reported number touches the full dataset.

6 Results

6.1 RQ1: Expert Agreement

²Panel consensus on Objective is constant across all 12 proposals, so Spearman ρ is undefined; the dimension is characterized by AC1 only.

Table 2: Per-dimension agreement against the expert panel ($n = 12$). AI columns compare the pipeline to panel consensus; human columns characterize the panel itself. LORO gives the range of each rater’s leave-one-out agreement with the other two.

Dimension	AI vs. panel		Panel internal		
	κ	ρ	ICC(2,1)	AC1	LORO ρ
Team	0.385	0.351	0.875	0.844	[0.86, 0.93]
Objective	0.000	— ²	−0.048	0.876	—
Strategy	0.333	0.386	0.794	0.778	[0.82, 0.91]
Advantages	0.226	0.147	0.789	0.784	[0.80, 0.91]
Feasibility	0.050	0.042	0.855	0.848	[0.89, 0.94]
Overall	—	0.05–0.17	0.859	—	[0.92, 0.95]

Panel reliability. The three experts agree substantially with one another, establishing the ceiling against which the pipeline is measured. Single-rater reliability is ICC(2,1) 0.79–0.88 on Team, Strategy, Advantages, and Feasibility, and 0.859 on the overall ranking; panel-mean reliability is ICC(2,k) 0.92–0.95 throughout (Table 2). Objective is the one dimension where variance-based coefficients mislead: ICC and κ sit near zero, yet Gwet’s AC1 is 0.876. The cause is marginal skew — the raters converge on a single score for almost every proposal, and consensus is constant at 4 across all 12 — which deflates κ and ICC even under near-total agreement. This is the standard κ paradox [?], and we report Objective through AC1 accordingly.

AI versus panel. Per-dimension agreement is weak (Table 2): weighted κ is modest on Team (0.385), Strategy (0.333), and Advantages (0.226), and effectively absent on Feasibility (0.050) and Objective (0.000). Spearman ρ spans 0.04–0.39, none significant at $n = 12$ [HOLM-CORRECTED p -VALUES TO FILL]. On the overall ranking, ICC(2,1)= 0.136 and Spearman $\rho \approx 0.05$ –0.17; the two end-points correspond to alternative rank-rounding conventions on the consensus scale, and we quote the range rather than a point estimate throughout (§8). Bootstrap intervals (Fig. 3) are correspondingly wide.

The pipeline falls below the human floor. The decisive comparison is not the absolute size of these correlations but their position relative to the panel’s internal agreement. Each rater’s leave-one-out agreement with the other two ranges from 0.80 to 0.95 across scorable targets, and on the overall ranking from 0.92 to 0.95. The pipeline’s agreement with consensus sits below the *weakest* human on *every* scorable target, by margins of 0.4–0.9 in ρ — not a marginal shortfall but an order of separation. Expert-alignment validity therefore fails at the level the audit was designed to test: whatever construct the pipeline is measuring, it is not the one on which three independent domain experts converge.

Ordering and verdicts. Pairwise rank concordance against consensus is 57.7%, close to the 50% expected from arbitrary ordering. Verdict accuracy is 83.3% (100% precision, 50% recall; TP= 2, FP= 0, FN= 2, TN= 8), but this figure should not be read as agreement: with only two GO predictions against a consensus GO set of four, and eight true negatives, the accuracy is carried by the base rate rather than by discrimination. A classifier predicting non-GO for every proposal would score 66.7% on the same data.

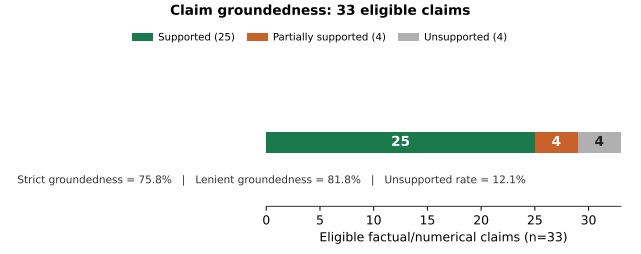


Figure 4: Distribution of the 33 eligible factual and numerical claims across support categories, with strict, lenient, and unsupported rates.

Table 3: Kickstarter outcome correlation ($n = 141$)

Group	n	Spearman ρ	Holm p
Toronto	104	0.314	0.007
Montreal	37	0.506	0.008
Combined	141	0.366	<0.001

Structured pipeline versus raw-document baseline. A baseline that scores directly off proposal text, with no fact-extraction or dimension-construction, provides a check on whether the nine-stage decomposition contributes measurable discrimination. [BASELINE METRICS AGAINST PANEL CONSENSUS TO FILL — Spearman ρ , ICC, distinct-value count, ordered pairs distinguished, tuned verdict accuracy.]

6.2 RQ2: Claim Groundedness

Across the 12 expert-scored proposals (§4.1), 188 report statements were classified: 114 professional inferences and 41 evaluator judgments (scores, confidence, verdicts) were excluded, leaving 33 eligible factual and numerical claims. Labels were model-assisted rather than independently double-annotated, a caveat we return to in §8. Of these, 25 were fully supported, 4 partially, and 4 unsupported (Fig. 4): strict groundedness 75.8% [59.0, 87.2], lenient 81.8% [65.6, 91.4], unsupported rate 12.1% [4.8, 27.3].

6.3 RQ3: Outcome Alignment and Threshold Transfer

The AI score correlates significantly with real funding outcomes across all three groups (Table 3). At the venture-calibrated threshold, the pipeline predicted GO for *zero* of 141 campaigns (0% recall), yet threshold-free discrimination remains strong (Fig. 5, ROC-AUC= 0.764). Stratified 5-fold cross-validated recalibration restores substantial recall (58.3%, F1= 0.364) versus an in-sample-only reference (F1= 0.478) that overstates performance — a measured illustration of overfitting risk in threshold calibration.

Per-dimension Spearman correlations, Holm-corrected across the five-test family: Feasibility is the strongest predictor of real outcomes ($\rho = 0.344$, Holm $p < 0.001$), followed by Strategy ($\rho = 0.252$, Holm $p = 0.010$) and Advantages ($\rho = 0.201$, Holm $p = 0.050$, at the significance boundary); Team ($\rho = 0.160$, Holm $p = 0.115$)

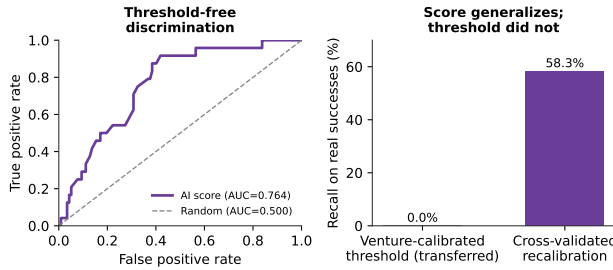


Figure 5: Left: ROC curve for the AI score against real Kickstarter funding outcomes (AUC=0.764, computed on all 141 proposals). Right: recall on real successes at the venture-calibrated threshold (transferred unmodified) versus after cross-validated recalibration.

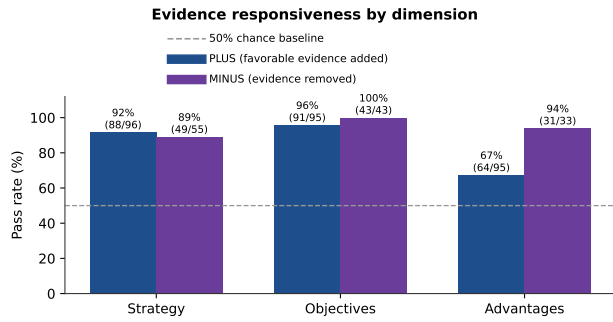


Figure 6: PLUS/MINUS pass rates by dimension. Advantages-PLUS is visibly the weakest cell, consistent with judgment-dependent sub-rubrics being harder to move than document-verifiable ones.

and Objective ($\rho = 0.088$, Holm $p = 0.302$) are not significant – the reverse of RQ1’s strongest predictor, Team.

6.4 RQ4: Evidence Responsiveness

Across the same 12 expert-scored proposals (§4.1), PLUS/MINUS perturbations moved scores in the expected direction on 366 of 417 checks (87.7% [83.3, 91.7], significant vs. a 50% baseline, $p < 0.001$). Each dimension is probed through eight fine-grained sub-rubrics: document-verifiable ones (milestone dates, revenue model, regulatory pathway, market size) tracked injected and removed evidence reliably, while judgment-dependent ones (problem-solution fit, market timing, competitive defensibility) were markedly less consistent, with Advantages-PLUS the weakest cell (67.4%, a medium effect, Cohen’s $h = 0.35$; Fig. 6).

7 Discussion

7.1 Dimension Importance Is Domain-Dependent

Team dominates the firm’s assessment (Table 2) but carries no significant signal for real Kickstarter funding (§6.3); Feasibility shows

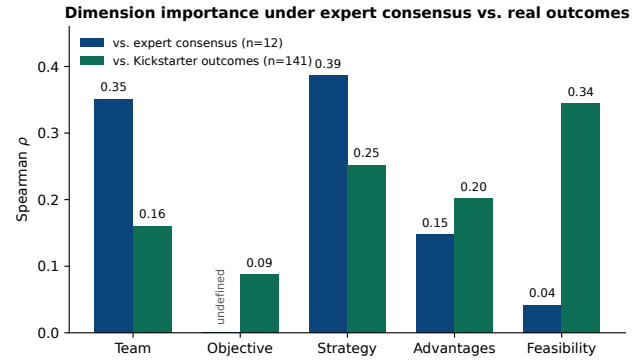


Figure 7: Dimension importance flips across domains: Team is the strongest predictor of the firm’s assessment (RQ1) but carries no significant signal for real Kickstarter outcomes (RQ3); Feasibility shows the reverse pattern.

the reverse (Fig. 7). The contrast is explicable rather than noise: venture investors weigh founder credentials because execution risk dominates early-stage failure, while backers cannot verify credentials and respond instead to whether the product looks buildable. The implication is that dimension weights validated in one financial domain encode that domain’s failure model and cannot be assumed to carry meaning elsewhere.

7.2 Score Generalizes; the Decision Threshold Does Not

A pipeline calibrated entirely on VC-style proposals produces a significant correlation with real crowdfunding outcomes it was never tuned on (Spearman $\rho = 0.366$, Holm $p < 0.001$; ROC-AUC=0.764) – yet its decision threshold, transferred unmodified, fires GO on zero of 141 campaigns (§6.3, Fig. 5). The mechanism is structural, not a scoring failure: the rubric’s score ceiling here (0.640 observed) sits below the transferred cutoff of 0.725, so no campaign can clear it regardless of quality. Recalibrating the identical OR-logic rule for Kickstarter, via held-out cross-validation, restores 58.3% recall – evidence this is a correctable threshold-transfer problem, not a broken score. The governance lesson: a score and its decision threshold carry separate validity, and confirming which failed requires outcome-based auditing.

7.3 Is Confidence a Governance Signal?

The pipeline emits a confidence value alongside every score, and confidence is one half of the OR-logic verdict gate (§3.4) – so its meaning matters operationally. On the 12-proposal study, neither of two checks validates it: confidence versus absolute score error is directionally correct but unreliable ($\rho = -0.413$, $p = 0.183$), and confidence versus verdict correctness shows essentially no relationship ($\rho = -0.048$, $p = 0.882$). Confidence should therefore be treated as an empirical decision feature, not a validated probability of correctness: a real number the system uses, but not shown to track whether the system is right. For human-supervised deployment, reviewers should not read high confidence as license to reduce scrutiny.

7.4 Implications for Governance and Human-Supervised Triage

Our findings translate into concrete deployment requirements. *Provenance and privacy*: minimize retained proposal text beyond what audit requires, restrict access to intermediate artifacts (raw facts, dimension evidence, candidate answers), and log the exact prompt and model version behind every recommendation. *Oversight*: support human override at every stage, and treat dimension-specific reliability as a routing signal – proposals turning heavily on Objective or Advantages should receive closer human review, given those dimensions' weaker measured agreement (Table 2). *Fairness*: mask or perturb demographic-adjacent cues (founder names, affiliations, nationalities) in Team scoring, and log separately when a judgment depends on affiliation rather than documented execution evidence. None of these safeguards are yet implemented; we report them as design targets, not delivered features. Together they define the boundary of our claim: the evidence supports auditable, human-supervised triage – not autonomous investment decisions.

8 Limitations

Statistical. At $n = 12$, correlation tests are underpowered ($|\rho| \approx 0.73$ required for 80% power), and the headline agreement result (Spearman $\rho = 0.589$) does not survive Holm correction ($p = 0.262$). No second rater exists, so no human-vs-human agreement ceiling can be established; the firm's file is an operational reference, not universal ground truth.

Methodological. The content-sensitivity checks are prompt-coupled, not independent tests of real-world responsiveness (§5.2). The groundedness audit (33 eligible claims) is a pilot with model-assisted, not double-annotated, labels. The rubric's target score band (§3.4) was set as a heuristic countermeasure to LLM central-tendency bias, not derived from this pool's true quality distribution; a systematically stronger or weaker pool may be biased toward the assumed band. Stage 6's commentary call runs at temperature 0.25 with untested run-to-run repeatability, and all results use a single provider (gpt-4o-mini).

Domain and fairness. Kickstarter precision after recalibration remains modest (26–33%) – sufficient for triage, not automated decisions – and composite-score improvements (§5.4) are directionally suggestive, not conventionally significant. Outcomes also depend on factors outside the proposal text (video production, social following, marketing), capping text-only precision regardless of calibration. Finally, Team scoring could leak demographic-coded signal from founder names and affiliations into execution-capacity judgments – a fairness risk we flag but have not audited; no masking is implemented (§7.4).

9 Conclusion and Future Work

We audited an evidence-grounded, nine-stage LLM venture-screening pipeline against expert judgment on 12 proposals and real funding outcomes from 141 Kickstarter campaigns. The central finding is that useful ranking and outcome signal can coexist with grounding, calibration, and cross-domain reliability failures. Although the score correlates significantly with previously unseen outcomes (Spearman $\rho = 0.366$, Holm $p < 0.001$; ROC-AUC = 0.764), its

decision threshold fails completely out-of-domain (0% recall). Expert agreement does not survive multiple-testing correction, scores show systematic downward bias, and confidence does not track correctness. These results shift the question from whether an LLM can produce a plausible investment score to what validation and governance are required before that score supports accountable, human-supervised triage. They do not justify autonomous investment decisions or general venture-success prediction.

Four audits remain future work: run-to-run repeatability, style robustness, cross-dimension halo effects, and cross-provider agreement. The highest-priority next steps are adding a second human rater to estimate the human-agreement ceiling and auditing Team-dimension fairness using demographic-cue masking (§8). Because Kickstarter provides objective, resolved outcomes at scale, this form of outcome-based validation can be extended efficiently, offering a practical way to audit LLM-based financial screening instruments before long-horizon labels mature.

References

- [1] Jae Yoon Bae, Simon Malberg, Joyce Galang, Andre Retterath, and Georg Groh. 2026. DIALECTIC: A Multi-Agent System for Startup Evaluation. arXiv:2603.12274 [cs.MA] doi:10.48550/arXiv.2603.12274
- [2] Divij Chawla, Ashita Bhutada, Do Duc Anh, et al. 2025. Evaluating AI for Finance: Is AI Credible at Assessing Investment Risk Appetite? arXiv:2505.18953 [cs.AI] doi:10.48550/arXiv.2505.18953
- [3] Rick Chen, Joseph Ternasky, Afriyie Samuel Kwesi, et al. 2025. VCBench: Benchmarking LLMs in Venture Capital. arXiv:2509.14448 [cs.LG] doi:10.48550/arXiv.2509.14448
- [4] Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, et al. 2024. Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference. In *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*, Vol. 235. PMLR, Vienna, Austria, 8359–8388.
- [5] Shuangyan Deng, Haizhou Peng, Jiachen Xu, Rui Mao, Ciprian Doru Giurcaneanu, and Jiamou Liu. 2025. FinMR: A Knowledge-Intensive Multimodal Benchmark for Advanced Financial Reasoning. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*. ACM, Singapore, 168–176. doi:10.1145/3768292.3770365
- [6] Eva Eigner and Thorsten Händler. 2024. Determinants of LLM-assisted Decision-Making. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, Honolulu, HI, USA, 1–14. doi:10.1145/3613904.3642607
- [7] Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. 2024. RA-GAS: Automated Evaluation of Retrieval Augmented Generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*. Association for Computational Linguistics, St. Julians, Malta, 150–158. doi:10.18653/v1/2024.eacl-demo.16
- [8] Yunfan Gao, Yun Xiong, Xinyu Gao, et al. 2024. Retrieval-Augmented Generation for Large Language Models: A Survey. *Comput. Surveys* 56, 6 (2024), 1–35. doi:10.1145/3637529
- [9] Jiahui Geng, Fengyu Cai, Yuxia Wang, Heinz Koeppel, Preslav Nakov, and Iryna Gurevych. 2024. A Survey of Confidence Estimation and Calibration in Large Language Models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Association for Computational Linguistics, Mexico City, Mexico, 6577–6595. doi:10.18653/v1/2024.naacl-long.366
- [10] Paul A. Gompers, Will Gornall, Steven N. Kaplan, and Ilya A. Strebulaev. 2020. How Do Venture Capitalists Make Decisions? *Journal of Financial Economics* 135, 1 (2020), 169–190. doi:10.1016/j.jfineco.2019.06.011
- [11] Sture Holm. 1979. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics* 6, 2 (1979), 65–70.
- [12] YongJoon Huh and Yeongeun Seo. 2025. Multilingual BERT-based Classification and Recommendation Model for Supporting Innovation Finance Decisions. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*. ACM, Singapore, 823–828. doi:10.1145/3768292.3770384
- [13] Seungone Kim, Jamin Shin, Yejin Cho, et al. 2023. Prometheus: Inducing Fine-Grained Evaluation Capability in Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Singapore, 14485–14512. doi:10.18653/v1/2023.emnlp-main.889
- [14] Seungone Kim, Juyoung Suk, Shayne Longpre, et al. 2024. Prometheus 2: An Open Source Language Model Specialized in Evaluating Other Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational*

- Linguistics (ACL 2024)*. Association for Computational Linguistics, Bangkok, Thailand, 6294–6315. doi:10.18653/v1/2024.acl-long.348
- [15] Md Tahmid Rahman Laskar, Sawzan Alqahtani, M Saiful Bari, et al. 2024. A Systematic Survey and Critical Review on Evaluating Large Language Models: Challenges, Limitations, and Recommendations. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Miami, Florida, USA, 13785–13816. doi:10.18653/v1/2024.emnlp-main.764
 - [16] Hoyoung Lee, Junhyuk Seo, Suhwan Park, et al. 2025. Your AI, Not Your View: The Bias of LLMs in Investment Analysis. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*. ACM, Singapore, 150–158. doi:10.1145/3768292.3770375
 - [17] Joohyun Lee and Minji Roh. 2024. Multi-Reranker: Maximizing Performance of Retrieval-Augmented Generation in the FinanceRAG Challenge. In *Proceedings of the Joint Workshop of the 8th Financial Narrative Processing and 5th Economics and NLP (FNP-ECONLP) @ EMNLP 2024*. Association for Computational Linguistics, Miami, Florida, USA, 133–139. doi:10.18653/v1/2024.finnlp-1.14
 - [18] Yuxuan Lei, Dingkang Yang, Jiawei Chen, Shunli Wang, Jing Liu, and Lihua Zhang. 2023. CFBenchmark: Chinese Financial Assistant Benchmark for Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Singapore, 1653–1670. doi:10.18653/v1/2023.emnlp-main.104
 - [19] Zhuoyan Li, Hangxiao Zhu, Zhuoran Lu, Ziang Xiao, and Ming Yin. 2025. From Text to Trust: Empowering AI-assisted Decision Making with Adaptive LLM-powered Analysis. arXiv:2502.11919 [cs.HC] doi:10.48550/arXiv.2502.11919
 - [20] Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Singapore, 2511–2522. doi:10.18653/v1/2023.emnlp-main.153
 - [21] Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. 2023. SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Singapore, 9004–9017. doi:10.18653/v1/2023.emnlp-main.557
 - [22] Sewon Min, Kalpesh Krishna, Xinxi Lyu, et al. 2023. FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Singapore, 12076–12100. doi:10.18653/v1/2023.emnlp-main.741
 - [23] Ekin Ozince and Yigit Ihlamur. 2024. Automating Venture Capital: Founder Assessment Using LLM-Powered Segmentation, Feature Engineering and Automated Labeling Techniques. arXiv:2407.04885 [cs.CL] doi:10.48550/arXiv.2407.04885
 - [24] Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. 2023. ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Singapore, 11584–11606. doi:10.18653/v1/2023.emnlp-main.708
 - [25] Preetha Saha, Jingrao Lyu, Arnav Saxena, Tianjiao Zhao, and Dhagash Mehta. 2025. Large Language Model Agents for Investment Management: Foundations, Benchmarks, and Research Frontiers. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*. ACM, Singapore, 736–744. doi:10.1145/3768292.3770387
 - [26] Patrick E. Shrout and Joseph L. Fleiss. 1979. Intraclass Correlations: Uses in Assessing Rater Reliability. *Psychological Bulletin* 86, 2 (1979), 420–428.
 - [27] Rickard Stureborg, Dimitris Alikaniotis, and Yoshi Suhara. 2024. Large Language Models are Inconsistent and Biased Evaluators. *arXiv preprint arXiv:2405.01724* (2024). arXiv:2405.01724 [cs.CL] <https://arxiv.org/abs/2405.01724>
 - [28] Shijie Wu, Ozan Irsoy, Steven Lu, et al. 2023. BloombergGPT: A Large Language Model for Finance. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: Industry Track*. Association for Computational Linguistics, Singapore, 408–422. doi:10.18653/v1/2023.emnlp-industry.30
 - [29] Qianqian Xie, Weiguang Han, Zhengyu Chen, et al. 2024. FinBen: A Holistic Financial Benchmark for Large Language Models. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Datasets and Benchmarks Track*, Vol. 37. 95716–95743. doi:10.52202/079017-3033
 - [30] Qianqian Xie, Weiguang Han, Zhengyi Lai, Min Peng, and Jimin Huang. 2023. PIXIU: A Large Language Model, Instruction Data and Evaluation Benchmark for Finance. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Singapore, 1631–1652. doi:10.18653/v1/2023.emnlp-main.103
 - [31] Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. 2023. FinGPT: Democratizing Internet-scale Data for Financial Large Language Models. In *Proceedings of the Fourth ACM International Conference on AI in Finance (ICAIF '23)*. ACM, New York, NY, USA, 335–343. doi:10.1145/3604237.3626889
 - [32] Boyu Zhang, Hongyang Yang, Tianyu Zhou, Muhammad Ali Babar, and Xiao-Yang Liu. 2023. FinEval: A Chinese Financial Domain Knowledge Evaluation Benchmark for Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Singapore, 1607–1630. doi:10.18653/v1/2023.emnlp-main.102
 - [33] Mengao Zhang, Jiayu Fu, Tanya Warrier, Yuwen Wang, Tianhui Tan, and Ke-wei Huang. 2025. FAITH: A Framework for Assessing Intrinsic Tabular Hallucinations in Finance. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*. ACM, Singapore, 159–167. doi:10.1145/3768292.3770433
 - [34] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, et al. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, Vol. 36. 57747–57770.